

Parkinson's Disease Progression — Project Proposal & Execution Plan

Big Data & Deep Learning Capstone. Part 1 refines the original project brief into a precise, defensible proposal. Part 2 is the team's working execution plan. Replace bracketed placeholders with your own details.

How to use this: Share Part 1 with your supervisor for sign-off before writing code. Use Part 2 as a living checklist — update the risk register if new issues come up.

Part 1 — Refined Project Proposal

1. Objective

Model individual-level longitudinal progression of Parkinson's disease by fusing multimodal clinical, biomarker, and imaging-derived data from the PPMI cohort, using deep learning architectures suited to irregular longitudinal data, with rigorous leakage-safe evaluation. Extend to smartphone-derived (mPower) signals as a secondary objective if data access, quality, and time allow.

2. Research Questions

RQ1 (core): Can **[MDS-UPDRS Part III motor score]** at 12 and 24 months be predicted more accurately from a patient's full longitudinal multimodal visit history than from a single baseline visit alone?

RQ2 (core): Among early (feature-level), late (decision-level), and attention-based (learned) fusion of clinical, biomarker, and imaging-derived modalities, which strategy gives the best — and most stable across prediction horizons — performance?

RQ3 (optional extension): Does adding mPower smartphone-derived voice, tapping, and gait features as an auxiliary modality improve on the PPMI-only model, despite device heterogeneity and crowdsourced noise?

Week 1 team decision: confirm or replace the bracketed target in RQ1. MDS-UPDRS Part III (motor score) is recommended — it's the most consistently collected scale across PPMI visits and maps directly to clinically meaningful severity. Whatever you choose, it must be this specific, not "progression" in general.

3. Prediction Task Definition

- **Unit of prediction:** one patient, at one point in their visit history
- **Input:** all visits up to and including time t

- **Output:** target score at $t + 12$ months and $t + 24$ months (two horizons, evaluated separately)
- **Task type:** regression (primary); optional secondary framing as binary classification ("meaningful worsening" vs. "stable," using an established clinical threshold) for interpretability
- **Hard rule:** no visit after time t may contribute to any feature used to predict at time t . This is checked explicitly during data engineering, never assumed.

4. Data & Cohort

- **Primary:** PPMI — clinical/motor/non-motor assessments, biospecimen markers, imaging-derived measures (DAT-SPECT striatal binding ratios, structural MRI regional volumes)
- **Cohort:** de novo PD, prodromal, and healthy control arms — finalize exact inclusion criteria in Week 1 once data access is confirmed
- **Secondary (stretch):** mPower — voice, finger-tapping, gait, and tremor from smartphone sensors (~58,000 recordings, ~5,800 participants)
- **MVP scope:** imaging as *derived features*, not raw 3D volumes. Raw-volume 3D CNN work is a stretch goal only (Part 2, Phase 4).

5. Methodology

5.1 Pipeline overview

flowchart TD

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A[Raw PPMI tables: clinical, biomarker, imaging-derived] --> B[Spark |
B --> C[Patient-level train/val/test split]
C --> D[Clinical and biomarker encoder: GRU-D or time-aware Transform
C --> E[Imaging-derived encoder: FFN or CNN]
D --> F[Fusion layer: early, late, attention - compared head to head]
E --> F
F --> G[Prediction: target at +12 and +24 months]
G --> H[Explainability: SHAP, attention, Grad-CAM]
```

5.2 Big Data component — where Spark is actually load-bearing

PPMI alone (~1,000 multimodal patients) is not large by volume — don't pretend otherwise in the report. Spark earns its place in two specific jobs:

1. **Longitudinal feature engineering** across the full cohort's visit history: windowed/rolling statistics, time-since-baseline features, missing-visit masks — a genuine distributed job over all patients × all visits × all modalities.
2. **(If RQ3 is pursued) mPower raw-signal processing:** feature extraction from tens of thousands of raw voice/accelerometer recordings — genuinely high-volume, and the strongest justification for distributed processing in this project.

5.3 Train/val/test protocol

Split **by patient ID**, never by visit/row. Compute the split once, save it as a shared table (`patient_id` $\rightarrow$ `split`), and have every team member load that same file. Nobody recomputes their own split.

5.4 Baselines (built before any fusion work)

- Per-modality classical ML on the most recent visit only (cross-sectional baseline)
- Per-modality simple sequence model (plain LSTM) on the full visit history

These exist to prove the pipeline works end-to-end and to give a real yardstick for the fusion models later.

5.5 Modality encoders

- Clinical/biomarker time series $\rightarrow$ GRU-D or a time-embedded Transformer (built to handle irregular visit spacing directly, not assume uniform time steps)
- Imaging-derived features $\rightarrow$ feedforward network or small CNN
- *Stretch*: raw 3D imaging CNN, following Dentamaro et al.'s architecture — only after the above is solid

5.6 Fusion strategies (compared head-to-head)

Early (feature-level concatenation), late (decision-level ensembling), and attention-based (learned) fusion — all three implemented and benchmarked, not just one chosen by assumption.

5.7 Explainability

SHAP for tabular/derived-feature contributions; attention-weight visualization for the sequence/fusion model; Grad-CAM if a raw-imaging CNN is built.

5.8 Stability evaluation

Re-run the fusion comparison across both prediction horizons (12 vs. 24 months). If the winning fusion strategy changes between horizons, that instability is itself a real, reportable finding for RQ2 — not a bug to hide.

6. Evaluation Metrics

- **Regression**: MAE, RMSE, R^2
- **Secondary classification framing**: F1, AUROC, calibration
- **Comparing fusion strategies**: paired significance testing (e.g., bootstrap), not raw metric differences alone

7. Ethics & Data Governance

- Full compliance with the PPMI Data Use Agreement (and mPower/Synapse terms, if RQ3 is pursued)
- No attempt at participant re-identification; all reporting is aggregate/de-identified
- Non-commercial academic use only; outputs restricted to the capstone report and any approved academic publication
- Any demo built for the final presentation is an internal, non-deployed illustration — not a clinical tool, not a public product (see Deliverables)

8. Deliverables

In scope:

- Spark-based data engineering pipeline
- Baseline models + fusion models with full comparison results
- Explainability analysis
- Written report (methodology, results, discussion, limitations)
- Internal demo (Streamlit/Gradio) for the defense — local, not deployed

Explicitly out of scope: a deployed, public-facing web application. This is a research pipeline, not a product — confirm with your supervisor if your program's rubric says otherwise.

9. References

1. Dentamaro, V. et al. (2024). *Enhancing early Parkinson's disease detection through multimodal deep learning and explainable AI: insights from the PPMI database*. Scientific Reports.
 2. Junaid, M. et al. (2025). *Multitask Deep Learning for Predicting Parkinson's Progression and Depression From Multimodal Time Series Data*. IEEE Access.
 3. Botha, H. et al. (2026). *Longitudinal voice biomarker trajectory modelling for Parkinson's disease severity: domain-adaptive transfer learning on mPower real-world smartphone data*. Frontiers in Digital Health.
 4. Lim, W. et al. (2025). *Smartphone-derived multidomain features including voice, finger-tapping movement and gait aid early identification of Parkinson's disease*. NPJ Parkinson's Disease.
 5. Dritsas, E. et al. (2025). *Applying Machine Learning on Big Data With Apache Spark*. IEEE Access.
 6. Bot, B. M. et al. (2016). *The mPower study, Parkinson disease mobile data collected using ResearchKit*. Scientific Data. (*The foundational mPower dataset paper — describes the study design and data collection. Note: distinct from Botha et al. above, which analyzes the resulting data.*)
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Part 2 — Team Execution Plan

Assumes a ~14-week semester. Adjust week numbers to your actual deadline.

10. Team Roles at a Glance

Role	Name	Primary focus	Key tools
Data & Big Data Engineering Lead	[Name]	Spark pipeline, feature engineering, the shared patient-level split, evaluation harness	PySpark, Docker/Databricks CE
Time-Series & Clinical Modeling Lead	[Name]	Clinical/biomarker sequence encoder, RQ1	PyTorch, GRU-D / Transformers
Imaging, Fusion & Explainability Lead	[Name]	Imaging encoder, all fusion strategies, RQ2, XAI	PyTorch, SHAP, Grad-CAM

All three should understand the full pipeline well enough to defend any part of it — ownership avoids duplicated work, it doesn't mean the other two ignore that piece.

11. Phase-by-Phase Plan

Phase 1 — Foundations (Weeks 1–2)

Goal: Data access moving, infrastructure ready, literature synthesized, target locked.

- **Data Engineering Lead:** Complete PPMI registration/DUA — this is the urgent one, start it immediately; set up the Spark environment (Docker or Databricks Community Edition); set up the shared GitHub repo (`data/` , `notebooks/` , `src/` , `report/`); read the Dritsas et al. (Spark) paper and Bot et al. (the mPower origin paper — just reading, no registration needed, useful background for whoever eventually builds the Phase 7 pipeline); start exploring PPMI's data dictionary as soon as documentation is available. *mPower/Synapse registration itself is not urgent — nothing in the core pipeline needs it. Defer that until Phase 7, only if RQ3 is still in scope by then.*
- **Time-Series Lead:** Extract Junaid et al. and Botha et al. into the literature table (below); shortlist irregular-time-series architectures (GRU-D, time-aware Transformers).
- **Imaging/Fusion Lead:** Extract Dentamaro et al. and Lim et al.; check which imaging-derived variables PPMI provides pre-computed vs. what would need raw processing.

- **Joint:** Lock the exact RQ1 target (Section 2); decide the exact Big Data placement (Section 5.2); fill in Section 3's task definition together.
- **Deliverable:** Literature comparison table completed; this proposal finalized as a team; repo + environment ready; data access requests submitted.

Literature extraction template (one row per paper):

Paper	Cohort / N	Exact target	Fusion strategy	Architecture	Split protocol	Result	What they didn't handle

Phase 2 — Data Engineering (Weeks 2–4)

Goal: Clean, joined, leakage-safe modeling tables.

- **Data Engineering Lead (owns this phase):** Build the Spark pipeline — ingest raw PPMI exports, join clinical/biomarker/imaging-derived tables by patient + visit, engineer time-respecting rolling/windowed features, handle missing visits (masks, time-delta features for GRU-D), create the canonical patient-level split and save it as the shared artifact.
- **Time-Series Lead:** Define the clinical/biomarker feature list with the Data Lead; build the time-series data loader; run EDA on visit frequency, missingness patterns, and target-score distributions.
- **Imaging/Fusion Lead:** Extract the specific imaging-derived variables to use; if raw imaging is a later stretch goal, research the preprocessing tooling now (don't discover this in week 10).
- **Joint:** Review the split and EDA together; freeze the feature list per modality before modeling starts.
- **Deliverable:** Clean modeling tables per modality + saved patient-level split + EDA summary.

Phase 3 — Baselines (Weeks 4–6)

Goal: Simple, working, single-modality baselines — the pipeline proven end-to-end, and a real yardstick for later.

- **Data Engineering Lead:** Build the shared evaluation harness (given predictions + targets, computes MAE/RMSE/R² and produces the comparison table everyone uses downstream).
- **Time-Series Lead:** Train a cross-sectional baseline (XGBoost/LightGBM on the latest visit only) and a longitudinal baseline (plain LSTM on the full clinical/biomarker history).

- **Imaging/Fusion Lead:** Train a classical ML or simple NN baseline on imaging-derived features alone.
- **Joint:** Compare all baselines in the shared harness; check for anything suspiciously perfect (leakage) or suspiciously random.
- **Deliverable:** Baseline comparison table — a good candidate for a mid-semester supervisor check-in.

Phase 4 — Modality-Specific Deep Encoders (Weeks 6–9)

Goal: A real deep encoder per modality, each beating its own baseline.

- **Time-Series Lead (owns):** Implement GRU-D or a time-embedded Transformer; tune against the Phase 3 LSTM baseline.
- **Imaging/Fusion Lead (owns):** Implement the imaging-derived feature encoder; only after this is solid, prototype a raw-imaging 3D CNN as a stretch add-on.
- **Data Engineering Lead:** Support both with pipeline adjustments as needed; if RQ3 is still in scope, start the mPower raw-signal Spark pipeline now, since it's the most time-consuming Big Data piece.
- **Joint:** Short weekly sync to catch integration issues (ID format mismatches, inconsistent scaling) before fusion time.
- **Deliverable:** Two validated modality encoders, each beating its Phase 3 baseline (or a documented reason why not).

Phase 5 — Fusion & Comparison (Weeks 9–11)

Goal: Answer RQ2 rigorously.

- **Imaging/Fusion Lead (owns):** Implement early, late, and attention-based fusion combining both encoders.
- **Time-Series Lead:** Support the clinical/biomarker side of fusion; help run the multi-horizon (12 vs. 24 month) evaluation.
- **Data Engineering Lead:** Run paired significance testing between fusion strategies and baselines using the shared harness.
- **Joint:** Interpret results honestly — a marginal fusion gain over the best single modality is a legitimate, expected finding, not a failure.
- **Deliverable:** Full fusion comparison table + significance tests — RQ2 answered.

Phase 6 — Explainability & Stability (Weeks 11–12)

Goal: Interpret the winning model; verify the stability claim in RQ2.

- **Imaging/Fusion Lead:** SHAP for feature contributions; Grad-CAM if a raw-imaging CNN was built.

- **Time-Series Lead:** Attention-weight visualization — which visits/timepoints mattered most.
- **Data Engineering Lead:** Re-run the fusion comparison across both horizons (and ideally multiple seeds) to produce the stability evidence.
- **Deliverable:** Explainability figures + stability results table.

Phase 7 — mPower / RQ3 (Stretch — only if ahead of schedule)

- **Data Engineering Lead:** Finish/scale the Spark-based mPower feature extraction (voice, tapping, gait).
- **Whoever has capacity:** Domain-adaptive integration of mPower as an auxiliary modality, following Botha et al.'s general direction.
- **Deliverable:** RQ3 answered — does mPower add signal despite the noise?

Phase 8 — Report, Demo & Presentation (Weeks 12–14, writing threaded throughout)

- **All three:** draft your own methodology subsection as you finish that piece — not all in the last week.
- **Data Engineering Lead:** Compiles the report structure; writes the Big Data pipeline and ethics sections.
- **Time-Series Lead:** Writes the RQ1 results section.
- **Imaging/Fusion Lead:** Writes the RQ2 (and RQ3 if applicable) results and explainability section; builds the Streamlit/Gradio internal demo (2–3 days, local only, not deployed).
- **Joint:** Final polish; rehearse the defense together so all three can answer questions about the whole pipeline, not just their own piece.

12. Risk Register

Risk	Mitigation
PPMI/mPower access delayed	Phase 1 has non-data-dependent work queued (infra, literature) so nobody is blocked
Imaging too time-consuming	MVP uses derived features only; raw 3D CNN is a pure stretch goal
Data leakage from bad splitting	One shared patient-level split, created once in Phase 2, used by everyone
Fusion only marginally beats best single modality	Expected outcome — report it honestly as a finding, not a failure

Risk	Mitigation
Vague prediction target	Resolved in Phase 1 by locking one precise, specific target
Scope creep into a deployed "product"	Explicitly out of scope per Section 8

13. Milestones & Check-ins

- **End of Phase 1:** Proposal finalized, environment ready, access submitted
- **End of Phase 3:** Baseline results — good moment for a supervisor check-in
- **End of Phase 5:** Fusion results — RQ2 answered
- **End of Phase 8:** Final submission and defense